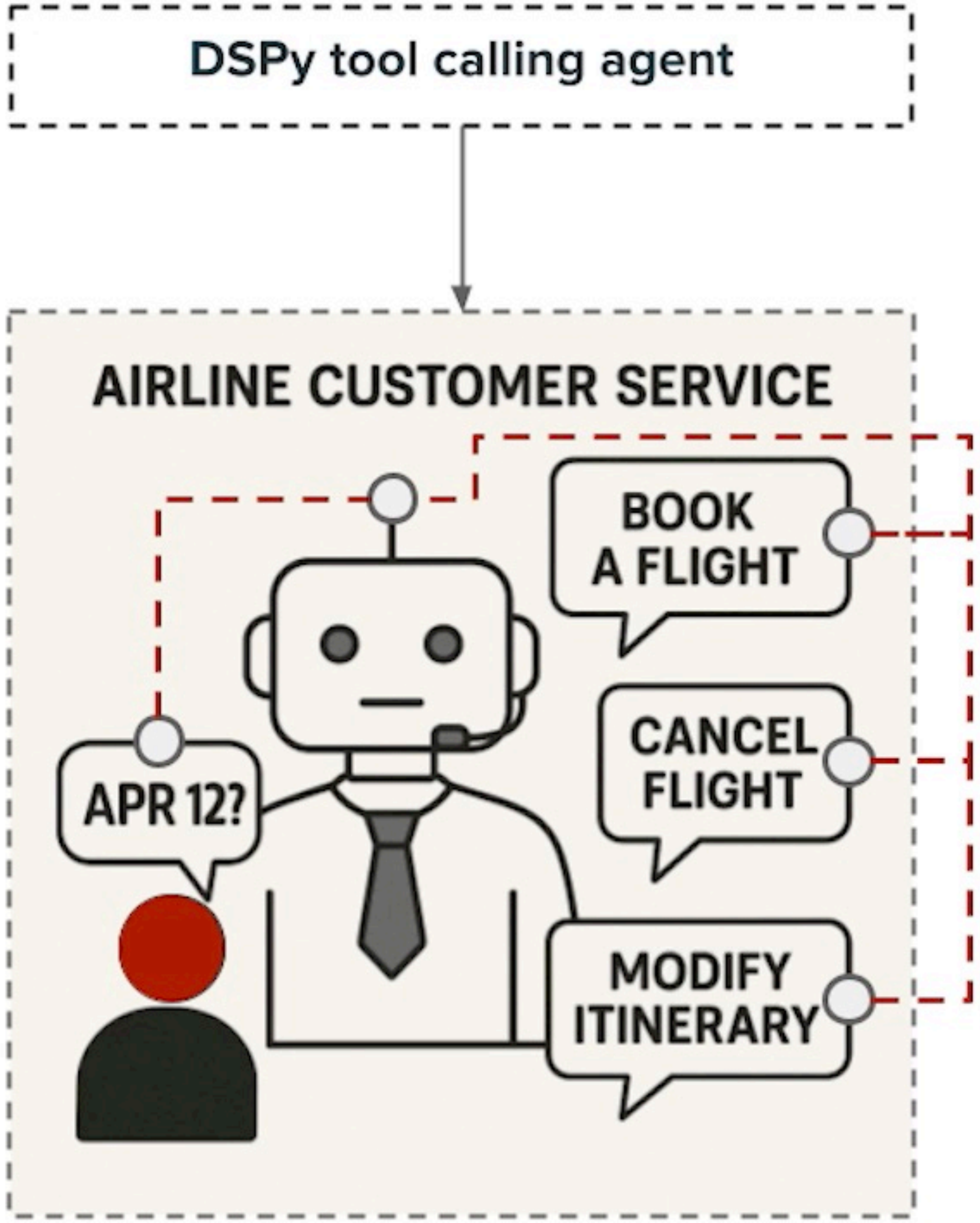
Use Managed MLflow Service

Databricks lighthouse: signup.databricks.com

Lab Materials



What is tracing, and why tracing?

- **Tracing:** Records **inputs and outputs** of intermediate steps and captures **hierarchical call stack**

- **GenAI apps are complex:** Internal logic is often opaque

- **Without tracing: One failed LM** call among many can **break the whole program**

Tracing improves:

✅ Interpretability

✅ Debugging

What is MLflow

Open source MLOps framework

- Supports **ML and GenAI app development**
- Covers the **full lifecycle**: development, tracking, deployment

Focus on reproducibility and traceability

- Manage every experiment, run, and model version with ease

Open source server + client

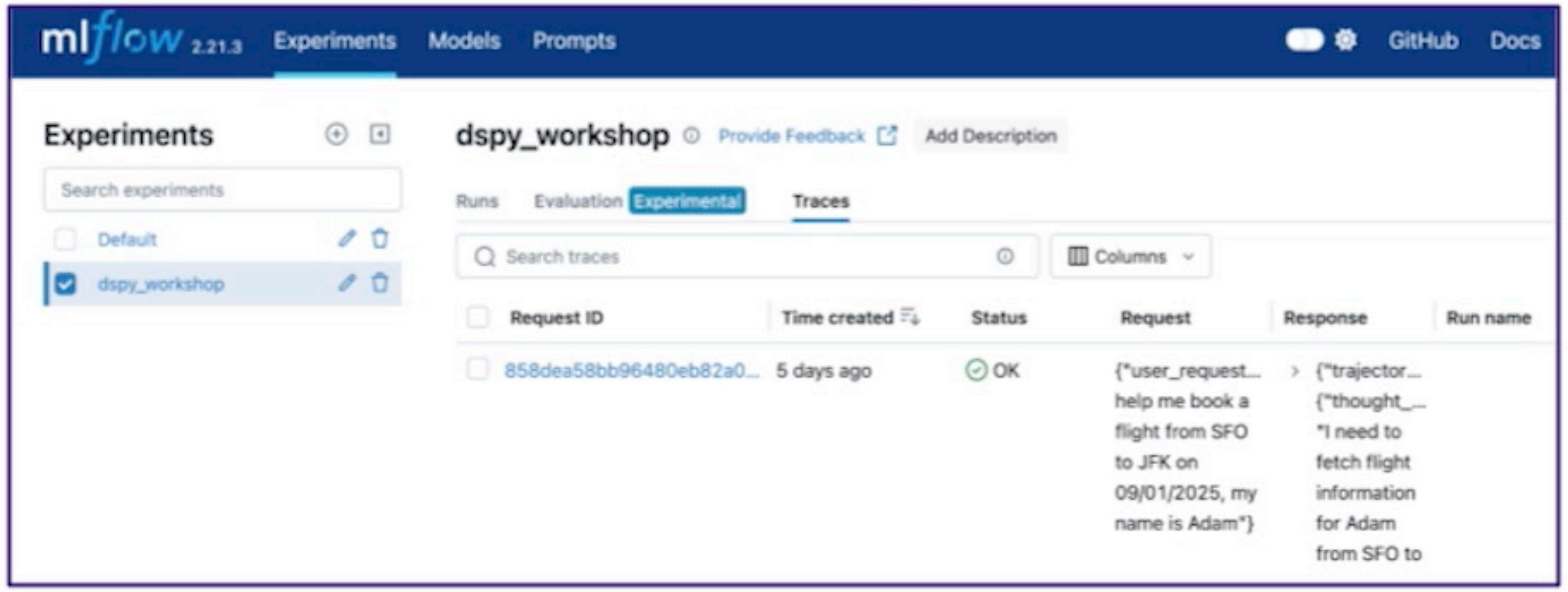
- Fully open and extensible

Get Started with Tracing DSPy

```python
import mlflow

mlflow.dspy.autolog()

class MyProgram(dspy.Module):
    def __init__(self, **kwargs):
        super().__init__()

        self.module1 = dspy.ChainOfThought(MySignature1)
        self.module2 = dspy.ChainOfThought(MySignature2)

    def forward(self, inputs, **kwargs):
        x = preprocess(x)
        x = self.module1(inputs)
        x = midprocess(x)
        x = self.module2(x)
        return postprocess(x)


my_program = MyProgram()

my_program(inputs="what's the meaning of life")
```

This is all you need!

MLflow Trace UI

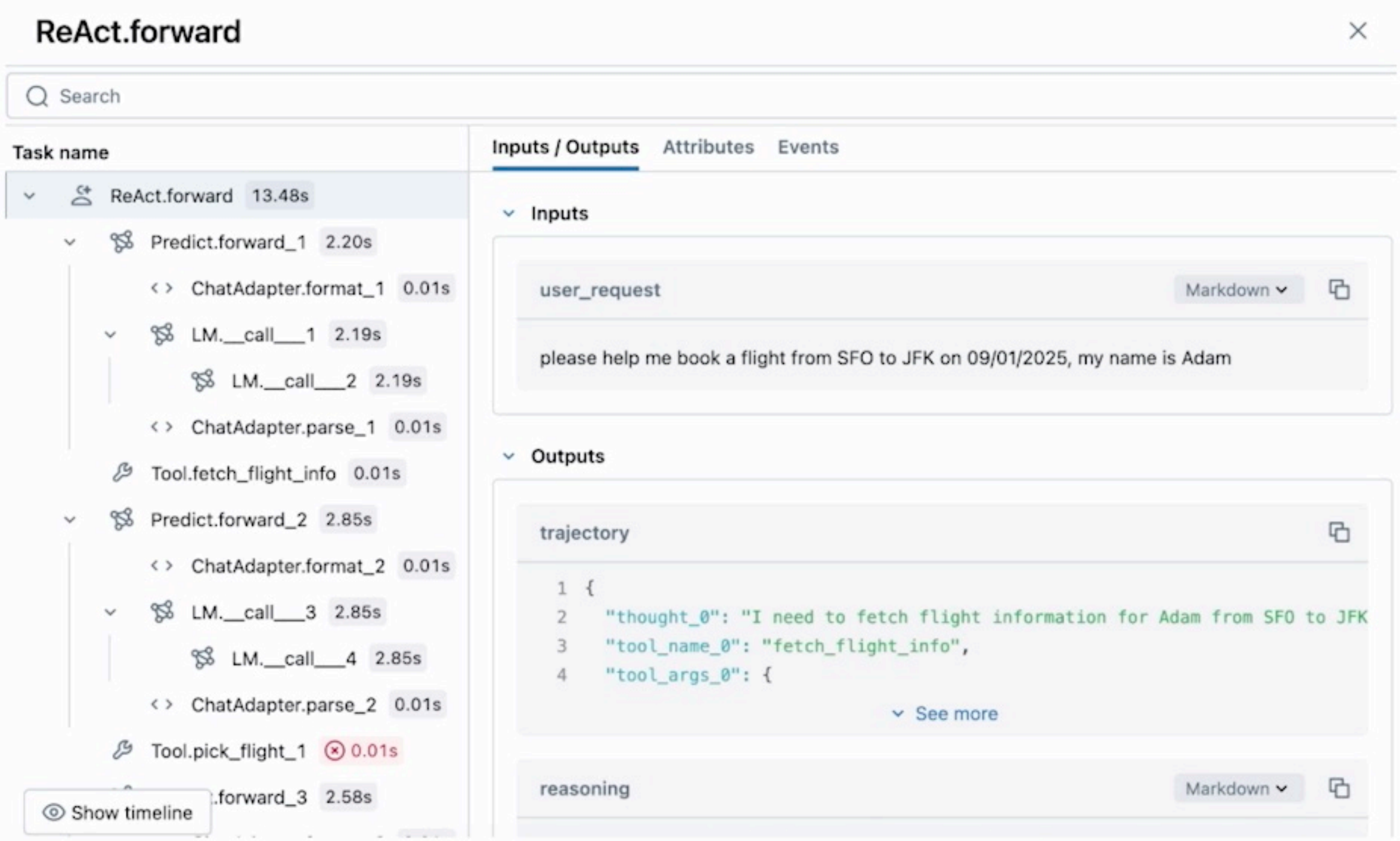


Traces are linked to MLflow experiments.

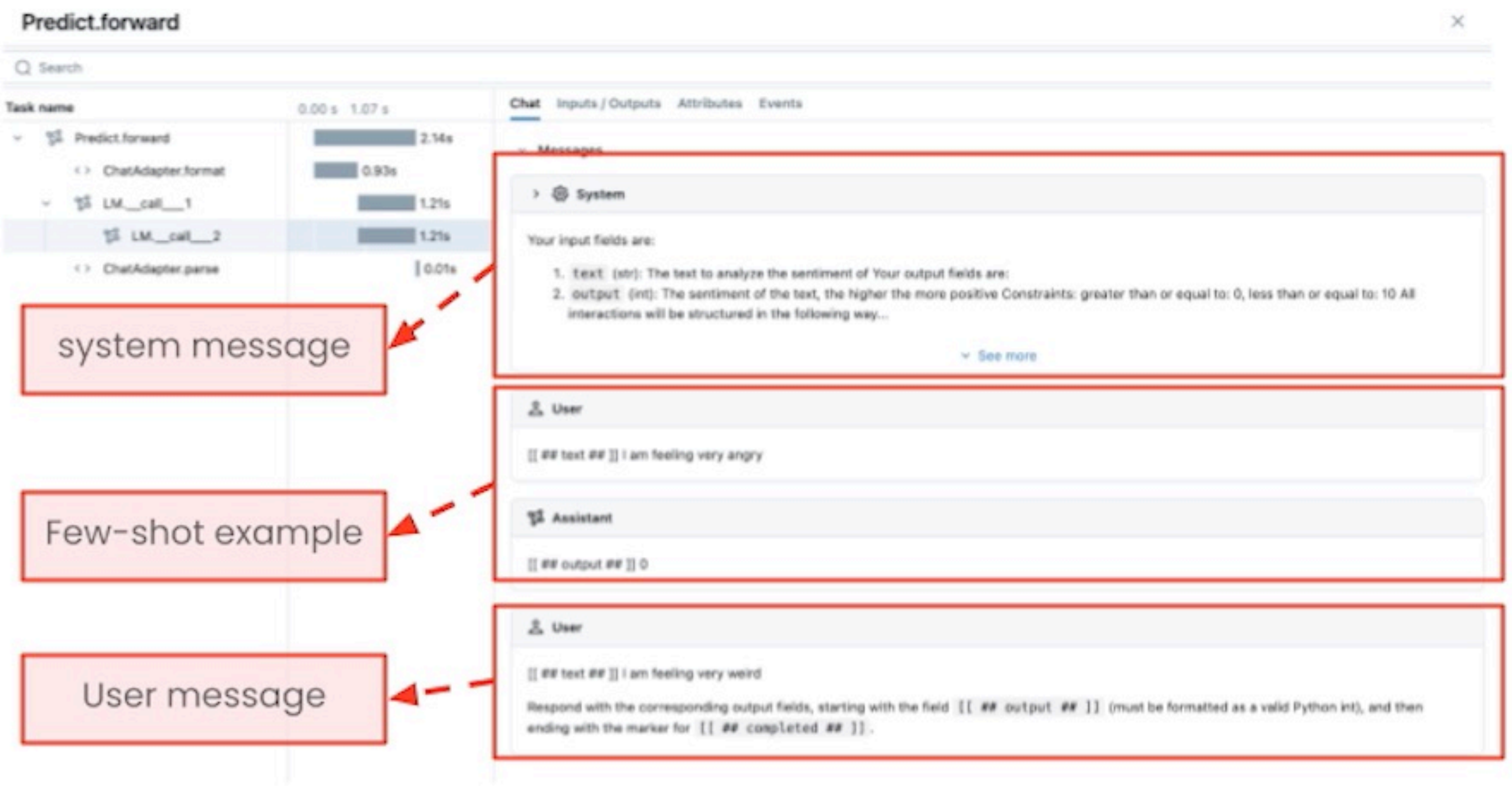
MLflow DSPy Trace Structure

The following DSPy components are traced:

- Module
- Adapter
- LM
- Tool



Use MLflow Trace to Interpret Prompt



Let’s see all of this in the lab